

Maintenance sheet version 10 for accumulation area (single mast, two levels)

Station name	SDM
People present	Ala, HETJA, BAV, FALO Johann & Greter
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	19 May 2025

Notes: ENVIRONMENT: ~10¹⁰/s, -7°C, sunny → OK for both work & take-off

AWS to do list		Check after maintenance			
Part	Shipped? (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N				Clean? ☒ / N not clean 9/20/201
Iridium modem IMEI number	N				
Inclinometer/compass	N		Attached to the radiometer? Y / N		
Satellite antenna	N				
Wind sensor upper	N		Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Wind sensor lower	N	Y	Black box pointing same direction as the radiometer? ☒ / N		
Temperature / humidity assembly upper	Y (Lufft)	Y	Fan spins? ☒ / N Sounds, OK? ☒ / N wiring inside logger box was wrong		
Temperature / humidity assembly lower	Y (Lufft) Kaisalc		Fan spins? Y / N Sounds, OK? Y / N		
Sonic ranger upper	Y	Y	Old sensor with new membrane? Y / N		
Sonic ranger lower	Y	Y	Old sensor with new membrane? Y / N Kept showing "NAN"		
Thermistor string	PVC tube Ext. cable		Extended with 1 PVC-tube		
Solar panel	N	N	Clean? ☒ / N NB: is not effective when partially covered		
GPS antenna	N	N	At top of logger enclosure? Y / N		
Logger box ID	N		CR1000 or <u>CR1000X</u> ? ID number:		
Logger enclosure	N		Replaced (including everything inside)? Y / ☒ N New desiccant bags? Y / ☒ N		
Battery box	N		New box? Y / ☒ N New desiccant bags? Y / ☒ N battery box was retrieved and raised		
Mast extension inserted?	Y	Y x 2	How many inserted? How long? 2 x 1.5m = 3m Notes:		
Mast	N		New mast? Y / ☒ N New sensor boom? Y / ☒ N		

Lower Vaisala: No reading, was loose upon arrival
Lower Young Wind loose (and turning) upon arrival

power cable plug was taken off to retrieve power cable, then re-mounted. Without this move we could not have raised AWS as power extension cable (new) was too short (3m mast extension)

Station name	SDM
Date	19 MAY 2025

APA, HETJA, BAY, PALO

Johanna Evert

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	- 00:28	00:00
Download data to PC and/or change SD card (wait for green light)	(Y/N)	Y/(N)
Name of logger program	GCNETv3.2	GCNETv3.2
Screen dump of values in fast scan mode	(Y/N)	(Y/N)
Photos of mast, sensors, logger box wiring etc. (more is better)	(Y/N)	(Y/N)
Upper boom direction with radiometer relative to north	$^{\circ}$ true/magn.	$^{\circ}$ true/magn.
Lower boom perpendicular to upper boom. If not, measure.	(Y/N)	Y/N
Box on upper wind sensor exactly same direction as radiometer? If not, measure.	(Y/N)	Y/N
Box on lower wind sensor exactly same direction as radiometer? If not, measure.	lose & turning Y/N	Y/N
Upper temperature / humidity height (bottom of casing - surface)	249 cm	cm
Lower temperature / humidity height (bottom of casing - surface)	lose 120 cm	~ 415 cm
Upper sonic ranger height (bottom of casing - surface)	205 cm	cm
Lower sonic ranger height (bottom of casing - surface)	87 cm	405 cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	(Y/N)	Y/N

Notes: Height snowboard stake : ^{32cm out} -44 cm $\rightarrow$ DEP : 148 cm, 230 cm w/ bamboo pole attached
 Depth to snowboard : 205 cm
 Height of thermistor PVC : -26 cm $\rightarrow$ DEP : ~ 180 cm
^{bamboo} $\rightarrow$ bamboo : 44 cm out $\rightarrow$ bamboo may be ~ 80 cm extra
 Depth to battery box : 205 cm

lower Valera loose and shield tilted, probe hanging out, inside full of packed snow

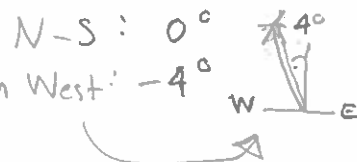
lower Young Wind sensor loose

Must-do at next visit:

inside logger box

Dept: Tilt of mast:

- The white wire on plug 4 (Temp/Hum/Fair/Luff) tilting down West: -4 $^{\circ}$
 we was placed into C1 and not C3, which was upper
 why the lufft didn't give any data.



SDM needs longer mast extensions (than 1.5m) - just like NSF & SDL

Bring enormously long power extension cable! The mast needs ~ 210cm raising at visits and the risk of not retrieving the battery box is high.